

DAPHNE CHEVALIER

M.Sc. in Plant Science | Biomedical Engineer

née Walford | walfordd@student.ubc.ca

HIGHLIGHTS

- Excellent communication skills in both English and French, with strong documentation habits
 - Experience writing, debugging, and documenting code with Python, R, MATLAB, C++, and C#
 - Strong organizational and collaborative skills developed via interdisciplinary projects with professionals and laypeople
 - Capable of executing complex projects, from a pediatric heart perfusion system to a tiny home
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EDUCATION

MSC. IN PLANT SCIENCE – UNIVERSITY OF BRITISH COLUMBIA 2022-2024

- Co-supervised by Dr. Juli Carrillo and Dr. Quentin Geissmann
- Designed and deployed multisensor stations to study effects of artificial light at night on insects
- Presented at Joint Annual Meeting of Entomological Societies of America, Canada, and British Columbia 2022, International Conference on Artificial Light at Night 2023, Entomological Society of British Columbia 2023 & 2024

UNCLASSIFIED STUDIES – UNIVERSITY OF BRITISH COLUMBIA 2020-2021

- For personal development, took a year of full-time studies on conservation, food systems, policy, and other topics

BASc, BIOMEDICAL ENGINEERING, HONOURS – UNIVERSITY OF WATERLOO 2015-2020

RESEARCH EXPERIENCE

Technician – Plant-Insect Ecology & Evolution Lab – Vancouver, Canada Jan 2025-present

- Currently conducting a field study to assess bias in insect biodiversity data collection methods
- Developed a Torch-based model in Python to automatically detect invasive insects in images

Team lead – HeartAgain Pediatric Transplant Heart Support System Nov 2018-Apr 2020

- *Project:* Keep pediatric transplant heart beating and monitored outside body, as part of biomedical engineering capstone design project at University of Waterloo
- Led team of 4 students in collaboration with Toronto's Hospital for Sick Children

Research Student – Department Of National Defence – Toronto, Canada Jan-Aug 2019

- *Project:* Developing a treatment to stop bleeding during non-compressible hemorrhage
- Optimised novel hemostatic agent with ELISA, gel electrophoresis, and thromboelastometry tests
- Results presented at Military Health System Research Symposium 2019 & published in two Defense Canada reports

Imaging Research Assistant – Sunnybrook Research Institute – Toronto, Canada May-Aug 2018

- *Project:* Improving accuracy of endovascular surgery with a steerable guidewire
- Redesigned interface of guidewire (now commercialised as the Magellan Cath-Pilot) in MATLAB

Research Assistant – Composite Biomaterial Systems Lab – Waterloo, Canada May-Aug 2016

- *Project:* Freeze-casting a novel material to create synthetic bone grafts
- Built and validated freeze-caster: machined parts and characterised cooling behaviour

PROFESSIONAL EXPERIENCE

Medical Office Assistant – SaltSpring Island Health Centre – Ganges, Canada May 2021-Apr 2022

- Coordinated appointments, maintained patient records, followed up on referrals and tests
- Helped patients feel calm and cared-for by listening supportively and directing them to resources
- Improved clinic website: created forms and pages for COVID-19 info & clinic closures

Product Development Student – Cellscale Biomaterials Testing – Waterloo, Canada Sep-Dec 2017

- Designed and tested motor controller for a novel device that tests properties of biomaterials
- Analysed product performance and disseminated results to customers

Software Quality Assurance Developer – NexJ Health – Toronto, Canada Jan-Apr 2017

- Wrote automated test suites for population health management system
- Redesigned part of the system's user experience to improve communication in circle of care

PUBLICATIONS

The Journal of Heart and Lung Transplantation. *A Closed-Circuit Ex Vivo Perfusion System for Pediatric Solid Organ Transplantation*. Haller, C., **Walford, D.**, et al. (Volume 40, Issue 4, Supplement). Toronto, ON, Canada, 2021.

Defence Research and Development Canada. *Development of biomaterials for treating battlefield hemorrhage Part 1: self-propelling particles*. Peng, H. and **Walford, D** (DRDC-RDDC-2020-R135; online). Toronto, ON, 2020.

Defence Research and Development Canada. *Development of biomaterials for treating battlefield hemorrhage Part 2: self-propelling particle-loaded gauzes*. Peng, H. and **Walford, D** (DRDC-RDDC-2022-R135; online). Toronto, ON, 2022.

PROJECTS

Detecting Astroturfing By Plastic Waste Non-Profits Nov-Dec 2020

- Executed for Writing and Research in the Disciplines class at University of British Columbia
- Compared website copy of anti-pollution non-profit organisations receiving plastic industry support ("astroturfing" organisations) to that of grassroots organisations by writing a Python script to analyse readability and word choice

TapChat Communication Device Jan-Apr 2018

- Completed with 4 teammates for Biomedical Engineering Design class at University of Waterloo
- Co-created assistive device to help stroke patients communicate with small finger movements

VOLUNTEERING

Salt Spring Island Water Preservation Society Salt Spring Island, Canada | Nov 2021-Apr 2022

- Collected and analysed data to estimate groundwater availability and recharge potential

Salt Spring Island Mental Wellness Initiative Salt Spring Island, Canada | Nov 2021-Apr 2022

- Translated and summarised research on peer support and counselling for new local initiative

Salt Spring Island Community Services Salt Spring Island, Canada | Jun-Aug 2020

- Collected produce donations and distributed nutritional coupons at weekly farmer's market

Grand River Hospital Waterloo, Canada | Jan 2018-Apr 2020

- Cared for elderly patients at risk of delirium for 3 hours/week, providing friendly conversation, activities, and range-of-motion exercises

AWARDS

Natural Sciences & Engineering Research Council of Canada:

Michael Smith Foreign Study Supplement (\$6,000 CAD, 4 months) 2023

Canada Graduate Scholarships—Master's (\$17,500 CAD, 1 year) 2023

Experience Award (\$4,219 CAD, 4 months) 2017

Walter Metzger Memorial Award (\$3,000 CAD) 2016

Fred Groch Entrance Bursary Award (\$3,000 CAD) 2015

INTERESTS & HOBBIES

- DIY and construction (built a tiny home from a modified open-source plan)
- Knitting, mending, organising a darning circle and helping other people learn to repair their clothing
- Woodcarving & woodwork, drawing & painting, music (piano & synthesizer), graphic design
- Backcountry hiking and camping, species identification, ocean swimming